



Internship Report Form

Student-Intern's Name	:	Skyler Jason P. Banzon				
Name of HTE	:	Cubeworks Technology Consulting and Solutions, Inc.				
HTE Supervisor's Name	:	Elmer N. Canque (CIO)				
Report Period	From	:	May 16, 2026	To	:	June 19, 2026
Number of Hours Worked This Quarter	:	168				
Cumulative No. of Hours to Date	:	584				

Quarter Summary

No.	Tasks Performed	% Time Spent
1	Embedded Display Project: Conducted extensive research into embedded GUI tools, evaluating the LVGL ecosystem and UI editing applications with a primary focus on layout compilation in EEZ Studio to establish a dedicated smart display lineup for factory environments.	15%
2	Embedded Display Project: Set up the ESP32-P4-ETH microcontroller with an LCD display, integrating MQTT client protocols directly into the newly designed user interface to enable real-time telemetry rendering and network state synchronization for factory machinery.	18%
3	System Integration: Programmed isolated system interface logic bridging back-end MQTT production metrics with the mock-up LVGL smart factory display screens deployed on the physical ESP32-P4-ETH hardware loop.	18%
4	Smart City Mapping Project: Deployed a photogrammetry asset pipeline using Google Earth Map 3D Exporter to capture detailed aerial topography data of Cordova, refining and texturing the geographic meshes within Blender to provide an alternative 3D model for utility reference.	17%
5	Smart City Mapping Project: Established a satellite-to-3D visualization pipeline extracting geographic datasets through QGIS and importing them into Blender to sculpt accurate terrain reliefs, delivering completed 3D models intended as visual assets for Cordova's pipe-monitoring initiative.	17%
6	Client Hardware Assembly: Assisted in the physical assembly and hardware configuration of specialized pressure monitoring devices designated for a corporate client.	15%
TOTAL		100%



Additional Activities (other growth opportunities e.g., conferences, training, seminars etc.)

- None

List any holidays, personal days, and/or days missed this quarter:

- Observed national holidays during the reporting period: Eid'l Adha & Independence Day. Missed two days due to the school events such as thesis defense schedule and batch recollection

Quarter Analysis, Reflection, and Plan of Action

What experiences were particularly rewarding during this reporting period?

Hands-On Hardware Assembly: Gaining practical experience through the physical assembly and precision soldering of specialized pressure monitoring devices for a corporate client.

UI-to-Hardware Integration: Successfully bridging the software and hardware gap by deploying custom LVGL user interfaces directly onto the ESP32-P4-ETH hardware loop and establishing stable MQTT communication.

What experiences were particularly difficult during this reporting period?

Resource-Intensive 3D Rendering: Processing and rendering large satellite geographic datasets in Blender and QGIS caused severe bottlenecks, as local hardware struggled to keep up with the massive computational demands.

EEZ Studio Learning Curve: Navigating the ecosystem and learning the intricacies of layout compilation within EEZ Studio for the smart factory display lineup proved highly challenging and time-consuming.

(If Applicable) Other comments, items, or issues that you want your supervisor or adviser to be made aware of? What is your proposed plan of action to resolve this issue?

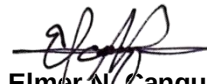
No major issues to report



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School of Engineering


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HTE Supervisor

Date Submitted : June 10, 2026

Date Reviewed : June 10, 2026

Supervisor's Comments:

None